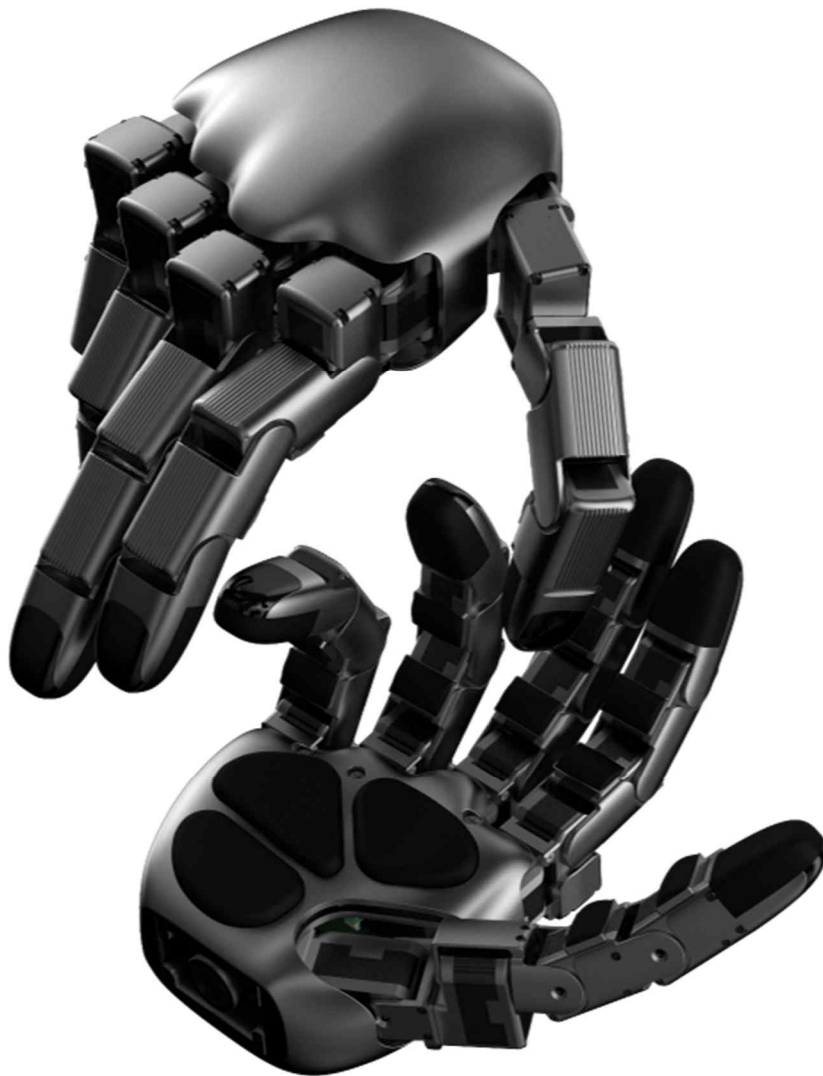




Allegro Hand V6 F

User's Manual



 **WONIK ROBOTICS**

Allegro Hand V6 F User's Manual

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1. In the Box

Included are the software and hardware necessary to get you started.

1. Allegro Hand	1 EA
2. Allegro Hand Case	1 EA
3. Power and Communication Cable (8 pin)	1 EA
4. Desktop Stand	1 EA
5. AC/DC Power Adapter	1 EA
6. Mounting Quick Changer	1 EA
7. Bolts for Mounting	
A. M4-8mm	4 EA
B. M6-10mm	4 EA

2. Allegro Hand Overview



Allegro Hand is a highly adaptive robotic hand.

With five fingers and twenty independent current-controlled joints and tactile sensors, it's the perfect platform for grasp and manipulation research.

Features

- Lightweight and portable anthropomorphic design
- Provides adduction and abduction capability of the fingers, enabling more natural and versatile grasping similar to a human hand
- Multiple ready-to-use grasping algorithms capable of handling a variety of object geometries
- 20 independent current-controlled joints (5 fingers x 4 DOF ea.)
- Equipped with 18 pressure sensing channels (5 fingertips, 10 joints, and 3 palm) to capture detailed force distribution for precise tactile feedback
- Support for real-time control and online simulation

3. Technical Specifications

Number of Fingers	4 fingers + 1 thumb = 5		
Degrees of Freedom	5 fingers x 4 = 20 (active)		
Actuation	Type	DC motor	
	Gear Ratio	288.35:1	
	Stall Torque	0.92 Nm	
	Nominal Torque	0.23 Nm	
Payload	Max. 15kg		
Weight	1,240g		
Joint Resolution	0.088 deg		
Communication	Interface	RS-485, Ethernet	
	Protocol	Modbus (RTU, TCP), RS-485 Custom	
Power Requirement	24.0 V / 5.0 A / 120 W		
Tactile Sensor *Pressure Sensor	Pressure Operating Range	260 to 1260 hPa (Atmospheric pressure is 1,013.25 hPa)	
	Color Indicator (Fingertips only)	Color	hPa Range
		 Blue	≤ 1050
		 Green	1050 ~ 1130
		 Olive	1130 ~ 1170
		 Red	≥ 1170
	Temperature Operating Range	0~60 °C	
	Pressure Accuracy	1 hPa	

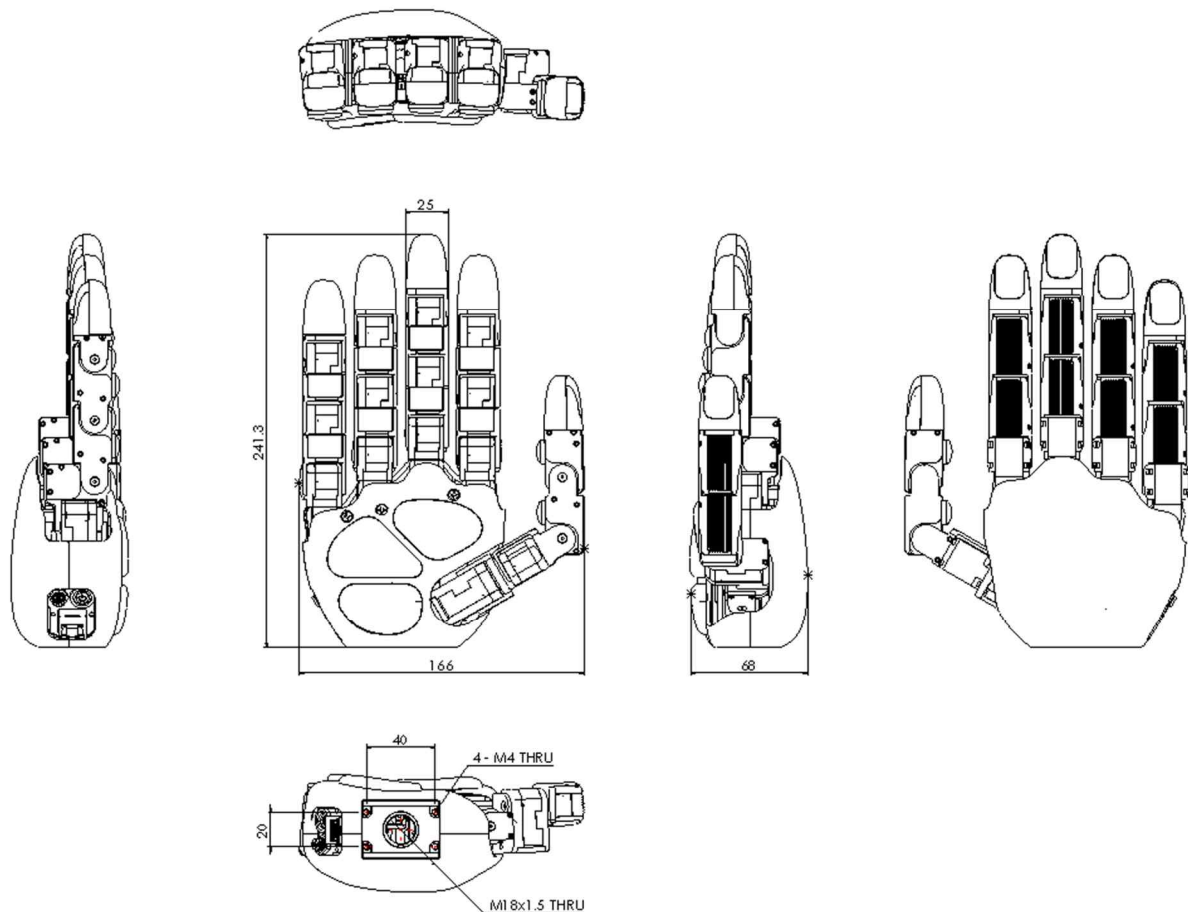


4. System Requirement

CPU	Intel® Core™ i3-8109U or higher
RAM	at least 2GB
Storage	at least 2GB
Graphics	OpenGL 3.0 H/W Acceleration enabled with at least 64Mb of video RAM
OS	Ubuntu® 22.04 LTS & 24.04 LTS
Additional S/W	ROS2

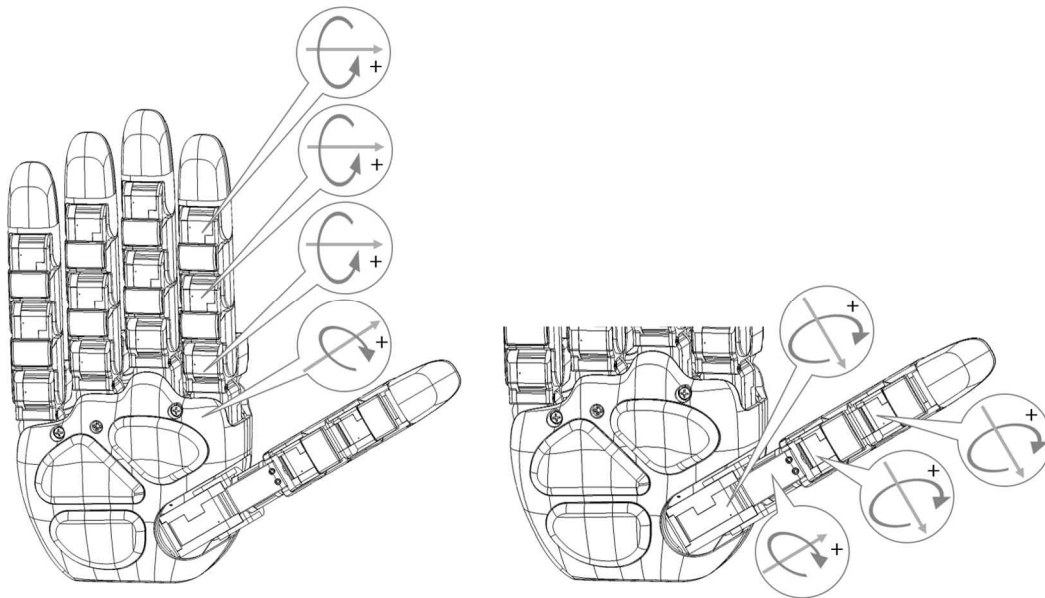
5. Joint Dimensions

All dimensions are displayed in millimeters (mm) and degrees.

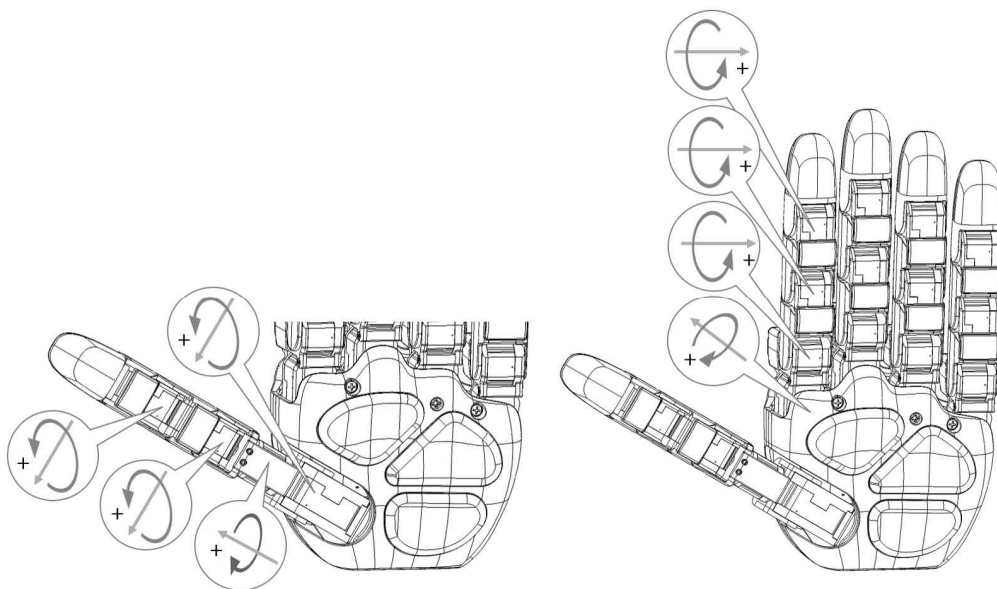


6. Joint Directions

6.1 Right Hand



6.2 Left Hand

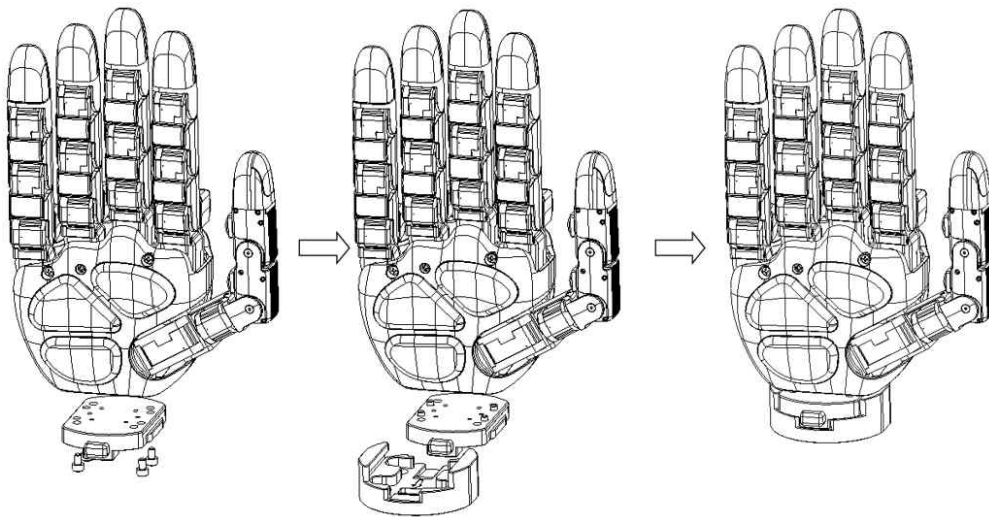


7. Mounting the Allegro Hand

All dimensions are displayed in millimeters (mm) and degrees.

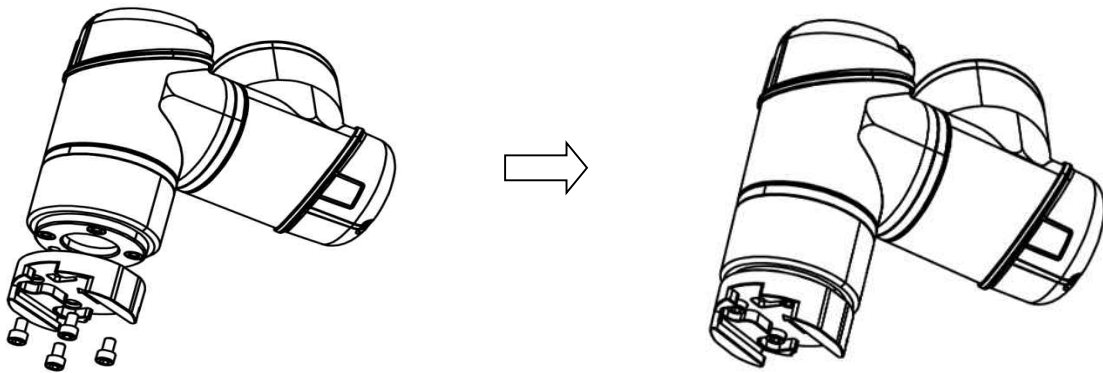
7.1 Mounting Block Removal

The mounting block is connected to the Allegro Hand using four M4x6mm hex socket head cap screws. Once the 4 screws have been removed, the mounting block can be removed from the bottom of the hand

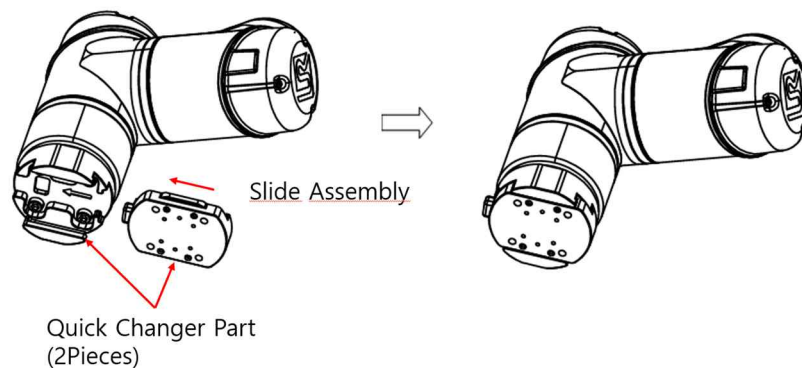


7.2 Mounting

The block can be mounted to the manipulator's end effector using four M6x10mm hex socket head cap screws.



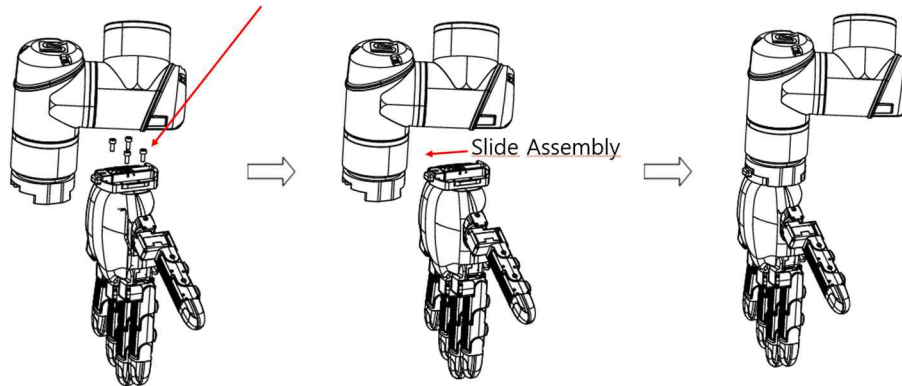
When changing the hand's orientation, please use a alternator bracket; please follow the instructions below.



7.3 Reassembly

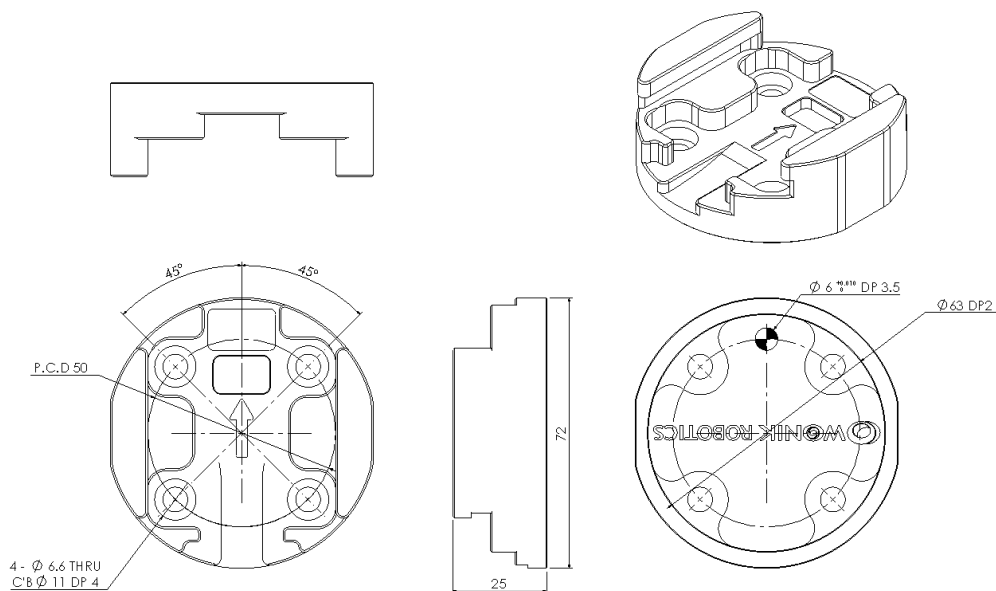
Place the hand onto the mounting block and replace the three M4x10mm hex socket head cap screws.

M4x10mm hex socket head cap screws.



7.4 Mounting Block Dimensions

The relevant dimensions of the mounting block are presented in millimeters (mm).



8. Allegro Hand Wiring

8.1 Wiring

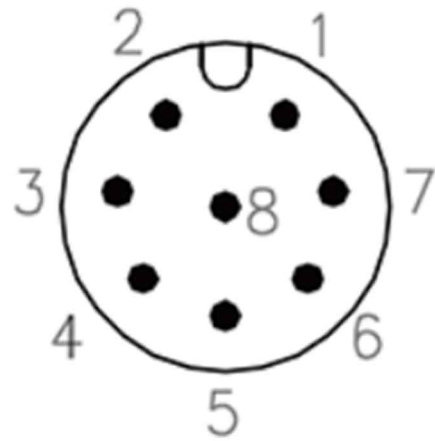
On the back of the Allegro Hand, you will see a connector. This connector is used to supply power to hand. It also has external interface. Connector used is CDM810-08A-MW-F810-050-67 part.

	Part Name	Note
Panel Mount Connector	CDM810-08A-MW-F810-050-67	M8 Stecker 8P, A-coding
Cable Connector	CDM818-08A-01FRA-2M-67	M8-F-Angle-8P-3m

Connector Pinout

No.	Description	Wiring Color
1	RS485-	White
2	RS485+	Brown
3	24VDC_IN	Green
4	GND	Yellow
5	24VDC_IN	Gray
6	GND	Pink
7	24VDC_IN	Blue
8	GND	Red

8-Pin Connector



8.2 Power

The Allegro Hand must be powered by DC 24V, 5A power supply. If a capacity of power supply is not sufficient, it will not work properly.

Once the power is supplied at the rated values, the power switch on the back of the Allegro Hand can be switched on to provide power to the hand.

9. Modbus Protocol

9.1 General Information

9.1.1 RS-485 Baud-Rate

Modbus RTU (RS-485) requires a baud rate of 2 Mbps or higher.

The default baud rate for RS-485 is set to **2 Mbps**.

9.1.2 Default Modbus-TCP Network Configurations

The default network configurations for the Allegro Hand to use Modbus TCP are as follows:

- **Default IP Address:** 192.168.1.100
- **Default GW Address:** 192.168.1.1
- **Default Subnet Mask:** 255.255.255.0
- **Port:** 520

[NOTE]

- The last byte (octet) of the MAC address is linked with the Slave ID.
- To apply the changes properly, **please reboot the hand (device) after configuration**. The previous settings will remain active until a reboot is performed.

9.2 Supported Modbus Function Codes

Function Code	Name	Target Area	Remarks
FC01 (0x01)	Read Coils	Coil Map (0x0000 ~ 0x0010)	Reads single/multiple coil status
FC03 (0x03)	Read Holding Registers	Register Map (0x0000 ~ 0x0310)	Reads single/multiple consecutive registers
FC05 (0x05)	Write Single Coil	Coil Map (0x0000 ~ 0x0001)	Controls a single coil (Factory-set items are write-protected)
FC15 (0x0F)	Write Multiple Coils	Coil Map (0x0000 ~ 0x0001)	Controls multiple coils
FC16 (0x10)	Write Multiple Registers	RW Register Block	Controls multiple registers
FC23 (0x17)	Read/Write Multiple Registers	RO & RW Register Block	Simultaneous write and read in a single packet

9.3 Modbus Register Map (Holding Register)

9.3.1 Read-Only (RO) Telemetry Block

- Supported Function Codes: FC03, FC23 (Fast Loop Feedback Reading)
- Description: Continuous reading range for sensor and system status data
- Address Ranges & Registers:
 - Slow Loop Read: 0x0000 ~ 0x0066 (Total 103 Registers)
 - Fast Loop Read: 0x0000 ~ 0x003D (Total 62 Registers)

Address (HEX)	Item Name	Data Type	Regs	Unit / Range	Description / Remarks
0x0000	f_pos[20]	int16_t	20	Raw	Fast Loop Joint Position (FC23/FC03)
0x0014	f_curr[20]	int16_t	20	Raw	Fast Loop Joint Current/Torque (FC23/FC03)
0x0028	imu_data[4]	int16_t	4	Raw	BNO055 IMU Data (Quaternion/Euler)
0x002C	hpa[18]	uint16_t	18	hPa	Pressure sensors (5 fingers + palm)
0x003E	<i>Reserved</i>	-	2	-	Padding
0x0040	f_temp[20]	uint16_t	20	°C	Motor Temperature (4 joints per finger)

0x0054	s_temp[18]	uint16_t	18	100 LSB/°C	Pressure Sensor Temperature
0x0066	joint_status	uint16_t	1	Bitmask	Joint Status (Fault, Limit, Error, etc.)
0x0067	<i>Reserved</i>	-	9	-	-
0x0070	fw_ver	uint16_t	1	Hex	Firmware Version (e.g., 0x0100 = v1.0)
0x0071	hand_type	uint16_t	1	Enum	0: Left Hand, 1: Right Hand

9.3.2 Read-Write (RW) Control & Configuration Block

- Supported Function Codes: FC03, FC06, FC16, FC23 (Fast Loop Control Writing)
- Description: Continuous writing and command control range for Fast Loop

[Fast Control Block]

Address (HEX)	Item Name	Data Type	Regs	Supported FC	Description / Remarks
0x0100	goal_pos[20]	int16_t	20	FC23, FC16, FC03	Fast Loop target position command (pulse)

0x0114	goal_curr[20]	int16_t	20	FC23, FC16, FC03	Fast Loop target current command (mA)
0x0128	op_mode	uint16_t	1	FC03, FC06, FC16	Joint Operating Mode (0: Current, 5: Pos) Default: 5, Real-time only (not saved to Flash)
0x0129	grasp_save_slot	uint16_t	1	FC03, FC06, FC16	Grasp position save trigger (1~5) Saves current position
0x012A	grasp_play_slot	uint16_t	1	FC03, FC06, FC16	Grasp position playback trigger (1~5) Sets goal_pos to saved position

[Gain & Profile Config Block]

Address (HEX)	Item Name	Data Type	Regs	Supported FC	Description / Remarks
0x0200	gain_kp[20]	uint16_t	20	FC03, FC16	Joint P Gain
0x0214	gain_ki[20]	uint16_t	20	FC03, FC16	Joint I Gain
0x0228	gain_kd[20]	uint16_t	20	FC03, FC16	Joint D Gain
0x023C	prof_vel[20]	uint32_t	40	FC03, FC16	Velocity Profile (2 Regs per joint)
0x0264	prof_acc[20]	uint32_t	40	FC03, FC16	Acceleration Profile (2 Regs per joint)

[System & Network Config Block]

Address (HEX)	Item Name	Data Type	Regs	Supported FC	Description / Remarks
0x0300	comm_sel	uint16_t	1	FC03	RS485 Comm Mode (RO) (1: Modbus, 0: Custom)
0x0301	slave_id	uint16_t	1	FC03, 06, 16	Modbus Slave ID (1 ~ 247)
0x0302	imu_config	uint16_t	1	FC03, 06, 16	BNO055 IMU Op Mode
0x0303	Reserved_0	uint16_t	1	FC03, 06, 16	RS485 Baudrate Config
0x0304	Reserved_1	uint16_t	1	FC03, 06, 16	(TBD) Development Option
0x0305	<i>Reserved</i>	-	11	-	Padding
0x0310	net_ip	uint32_t	2	FC03, 16	IPv4 Address

0x0312	net_mask	uint32_t	2	FC03, 16	Subnet Mask
0x0314	net_gw	uint32_t	2	FC03, 16	Gateway Address
0x0316	net_dns	uint32_t	2	FC03, 16	DNS Address
0x0318	serial_no	uint64_t	4	FC03	Serial Number

※ The comm_sel setting can be changed via the comm_sel pins on the switch block of the Allegro Hand's internal control board. By default, this is set to Modbus.

[CAUTION]

- When network settings (IP address, Subnet Mask, Gateway, DNS) or major communication parameters are modified, the new values are saved to the flash memory.
- To apply the changes properly, **please reboot the hand (device) after configuration**. The previous settings will remain active until a reboot is performed.

9.3.3 Coil Map (Digital Control Flags)

- Supported Function Codes: **FC01** (Read Coil), **FC05** (Write Single Coil), **FC15** (Write Multiple Coils)

Address (HEX)	Item Name	Data Type	Supported FC	Description / Remarks
0x0000	torque_enable	Bool	FC01, 05, 15	Servo Torque On/Off (0: OFF, 1: ON)
0x0001	joint_calib	Bool	FC01, 05, 15	Joint Calibration Start (0: Idle, 1: Start)

※ For detailed information, refer to 9.3.8 Position Calibration.

9.3.4 FC23 Fast Loop Operation Guide

The **FC23 (Read/Write Multiple Registers)** function code is specifically designed for high-performance **Fast Control Loops**. It allows the host PC (Master) to update target commands and retrieve real-time sensor/state feedback within a single communication cycle, significantly reducing latency compared to running separate write and read commands.

9.3.4.1 FC23 Request Parameter Specifications

When configuring an FC23 request packet from the Master to the **Allegro Hand**, use the following parameters:

- **Read Starting Address:** 0x0000 (Points to f_pos)
- **Read Quantity:** 40 Registers (f_pos [20 regs] + f_curr [20 regs])
- **Write Starting Address:** 0x0100 (Points to goal_pos)
- **Write Quantity:** 40 Registers (goal_pos [20 regs] + goal_curr [20 regs])
- **Write Byte Count:** 80 Bytes (40 registers $\times$ 2 bytes)

9.3.4.2 Transaction Flow & Execution Order

Single RTT (Round Trip Time) Exchange

A single packet transmission from the host PC updates goal_pos and goal_curr while simultaneously retrieving the latest joint positions (f_pos) and currents/torques (f_curr). This cuts communication latency by more than half compared to sequential FC16 (Write) and FC03 (Read) operations.

Allegro Hand Internal Execution Sequence

To maintain deterministic control performance, the **Allegro Hand** Modbus receive

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ISR/Task processes the packet in a strict internal sequence:

1. **Write Processing First:** The **Allegro Hand** immediately updates the internal target command registers (goal_pos and goal_curr) with the values provided in the request packet.
2. **Read Processing Follows:** Immediately after updating the targets, the **Allegro Hand** captures the latest sensor states and joint feedback.
3. **Response Transmission:** The **Allegro Hand** packages the feedback data into the response frame and sends it back to the host PC.

9.3.4.3 Implementation Recommendations for Host Developers

- **Control Loop Timing:** Design the host control loop to match or exceed the **Allegro Hand** target cycle to fully leverage the speed benefits of FC23.
- **Data Types:** Ensure that goal_pos, goal_curr, f_pos, and f_curr are handled as signed 16-bit integers (int16_t) in a contiguous memory block to prevent misalignment errors during packing and unpacking.

9.3.5 joint_status Bit Mask Configuration

Bit Range	Name	Description
Bit 0 ~ 3	HW_INT_ERR	Hardware Integration Error (Overload, Shock, Overheat, or Voltage). Triggered if any joint encounters an error.

Bit 4	F0_COMM_ERR	Finger 0 Communication Error (ID match/timeout failure)
Bit 5	F1_COMM_ERR	Finger 1 Communication Error (ID match/timeout failure)
Bit 6	F2_COMM_ERR	Finger 2 Communication Error (ID match/timeout failure)
Bit 7	F3_COMM_ERR	Finger 3 Communication Error (ID match/timeout failure)
Bit 8	F4_COMM_ERR	Finger 4 Communication Error (ID match/timeout failure)
Bit 9	F0_FAULT	Finger 0 Internal Fault Status
Bit 10	F1_FAULT	Finger 1 Internal Fault Status
Bit 11	F2_FAULT	Finger 2 Internal Fault Status
Bit 12	F3_FAULT	Finger 3 Internal Fault Status
Bit 13	F4_FAULT	Finger 4 Internal Fault Status
Bit 14 ~ 15	Reserved	Reserved for future expansion (Maintain as 0)

9.3.6 Finger & Pressure Sensor

Finger Index Reference

Finger Index	Finger Name	Description
0	Thumb (F0)	Thumb finger joints
1	Index (F1)	Index finger joints
2	Middle (F2)	Middle finger joints
3	Ring (F3)	Ring finger joints
4	Little (F4)	Little (Pinky) finger joints

Pressure Sensor Index Reference

For the pressure sensor array (hpa[18]), each finger is equipped with 3 sensors, indexed sequentially starting from the **fingertip**.

Sensor Index	Finger Position	Description
0	Fingertip (Distal)	Closest to the tip of the finger
1	Middle Phalange	Center segment of the finger
2	Proximal Phalange	Closest to the base of the finger



9.3.7 IMU Config & IMU Data

The Allegro Hand V6 is equipped with a BNO055 IMU sensor and supports the following configurations:

9.3.7.1 Sensor Operating Modes

The BNO055 sensor supports Raw Data modes, which output raw sensor measurements, and Sensor Fusion modes, which calculate 3D orientation via internal algorithms. The output data format follows the `imu_data` array specifications.

Raw Data Modes

Outputs physical sensor measurements directly without post-processing.

- **0x00 IMU Disable:** Disables the IMU sensor (**Factory Default**)
- **0x01 ACCONLY:** Outputs raw acceleration values ([accel_x, accel_y, accel_z, 0])
- **0x02 MAGONLY:** Outputs raw geomagnetic (magnetometer) values ([mag_x, mag_y, mag_z, 0])
- **0x03 GYROONLY:** Outputs raw gyroscope values ([gyro_x, gyro_y, gyro_z, 0])

Sensor Fusion Modes

Outputs 3D spatial orientation in quaternion format ([w, x, y, z]) through internal fusion algorithms.

- **0x08 IMU:** Accelerometer and gyroscope fusion (magnetometer unused, relative orientation)
- **0x09 COMPASS:** Heading fusion including magnetometer
- **0x0A M4G:** Magnetically-based fusion
- **0x0B NDOF_FMC_OFF:** 9-axis full fusion (fast magnetometer calibration feature disabled)
- **0x0C NDOF:** 9-axis full fusion (absolute orientation measurement)

9.3.7.2 Unit Scales

The default output data resolution and scale standards for the BNO055 sensor are as follows:

Data Type	Symbol / Name	Scale & Unit	Description
Acceleration (Raw)	RAW_ACCEL	1 LSB = 1 mg	Accelerometer raw data scale
Magnetometer (Raw)	RAW_MAG	1 LSB = 1/16 μ T	Magnetometer raw data scale
Gyroscope (Raw)	RAW_GYRO	1 LSB = 1/16 dps	Gyroscope raw data scale (Degrees Per Second)
Orientation (Quaternion)	QUATERNION	1 LSB = 1/16384	Fusion mode output quaternion scale (dimensionless)

9.3.8 Position Calibration

To set the current position as the origin (Home position), perform the following steps:

1. **Prerequisites:** Ensure that the torque_enable register (Coil Address 0x0000) is set to 0 (**OFF**).
2. **Trigger Calibration:** Set the joint_calib register (Address 0x0001) to 1 (**ON**).

3. **Finish Calibration:** Set the joint_calib register (Address 0x0001) back to 0 (OFF).
4. **Verification:** Monitor the f_pos register. Calibration is successfully completed when the value of f_pos changes to 0.

Warning: Ensure that the robotic hand is in the desired home position before initiating calibration, as this process overwrites the origin offset.

9.3.9 Grasp Position Storage & Playback

Grasp Position Storage & Playback (Slots 1~5)

Users can store and retrieve up to 5 custom grasp positions. Position data saved in these slots is stored in the Allegro Hand's internal **Flash memory (Non-volatile)**.

- **Saving a Grasp Position:**
 - Manually adjust the Allegro Hand fingers to the desired grasp position.
 - Write the desired slot number (1~5) to the grasp_save_slot register.
 - The Allegro Hand captures and stores the current joint positions into the selected Flash memory slot.

Important Note: Saving to the internal Flash memory requires **a few seconds**. Please ensure that no other communication commands are sent to the hand while the write process is in progress to prevent data corruption or save failures.

- **Playing a Grasp Position:**
 - Write the desired slot number (1~5) to the grasp_play_slot register.

- The Allegro Hand sets the goal_pos to the saved position and executes the grasp motion.
- **Execution Conditions:** To successfully perform the grasp motion, the following conditions must be met:
 - **Operating Mode:** Set to Pos Control mode.
 - **Torque Status:** torque_enable must be set to ON.
 - **Goal Current:** goal_curr must be set to a valid value.

10. Technical Support

The newest version of all documents found in this manual can be found in the Allegro Hand website at

<https://www.allegrohand.com/>

A forum is also available on the website where users can share problems and solutions with each other and with WONIK ROBOTICS.

For any further questions please contact us.

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11. FAQ

11.1 Hardware FAQ

1. CAD STL Files

If you purchase, we will provide STL files.

Please give us your Hand's serial number.

2. Communication interface

Modbus (RTU, TCP), RS-485 Custom communication is supported.

3. Fingertip sensor

The contact surface of the fingertip sensor is made of silicone, and it operates by measuring internal air pressure. If it is punctured by sharp objects such as an awl or knife, internal air may leak, which can cause malfunction.

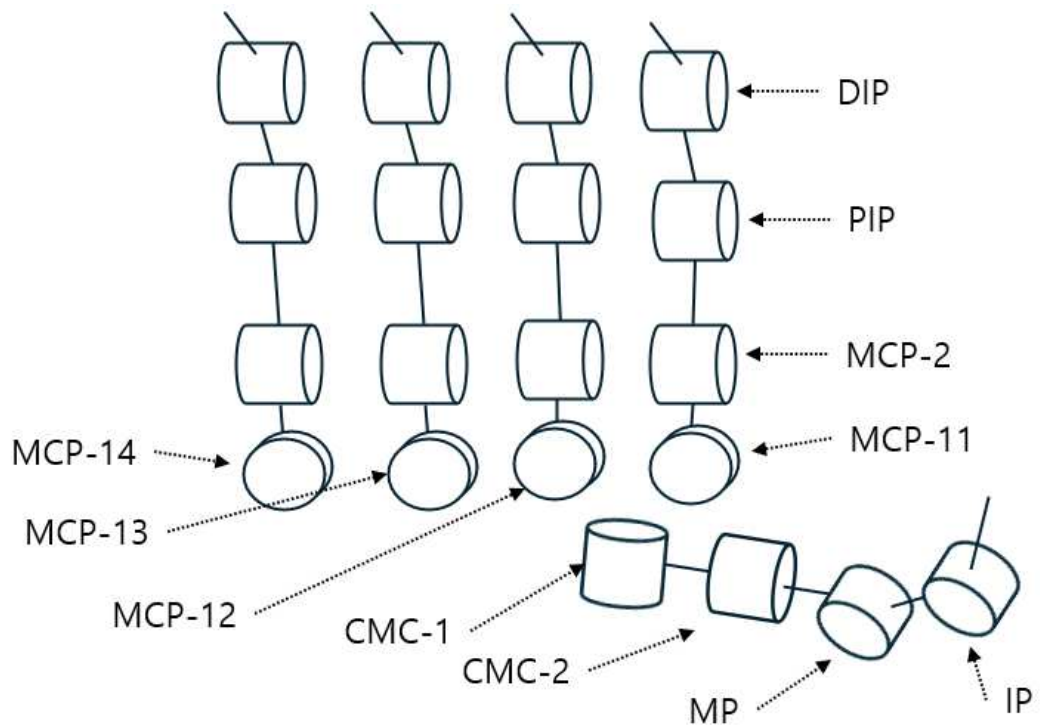
4. How robust is it to collisions?

Difficult to use in harsh environments.

It should be noted that this is for research purposes only.

Please avoid hitting a hammer or applying any excessive force.

5. Range of motion



	Joint	ROM [deg]
Finger	MCP-11	-22 ~ 75
	MCP-12	-65 ~ 65
	MCP-13	-75 ~ 22
	MCP-14	-75 ~ 25
	MCP-2	-4 ~ 90
	PIP	-10 ~ 80
	DIP	-10 ~ 80
Thumb	CMC-1	-2 ~ 95
	CMC-2	-95 ~ 95
	MP	-10 ~ 75
	IP	-10 ~ 80

11.2 Software FAQ

1. Is it possible to use V4, V5_Plus or V5_Sense software?

No.

There are many differences between V6 and previous versions.

2. How to set (left or right) of hand in SW?

Software automatically applies to the appropriate handedness and hand type through the serial number. Therefore, **you can simply use it as is.**

3. Where can PC S/W issues be supported?

You can use our official Forum: <https://www.allegrohand.com/forum>

Or you can check our official GitHub latest release, see the link below.

<https://github.com/Wonikrobotics-git>

Release Note

Date	Ver.	Description	Editor
2026.09.04	1.0	Initial draft	BK Lee